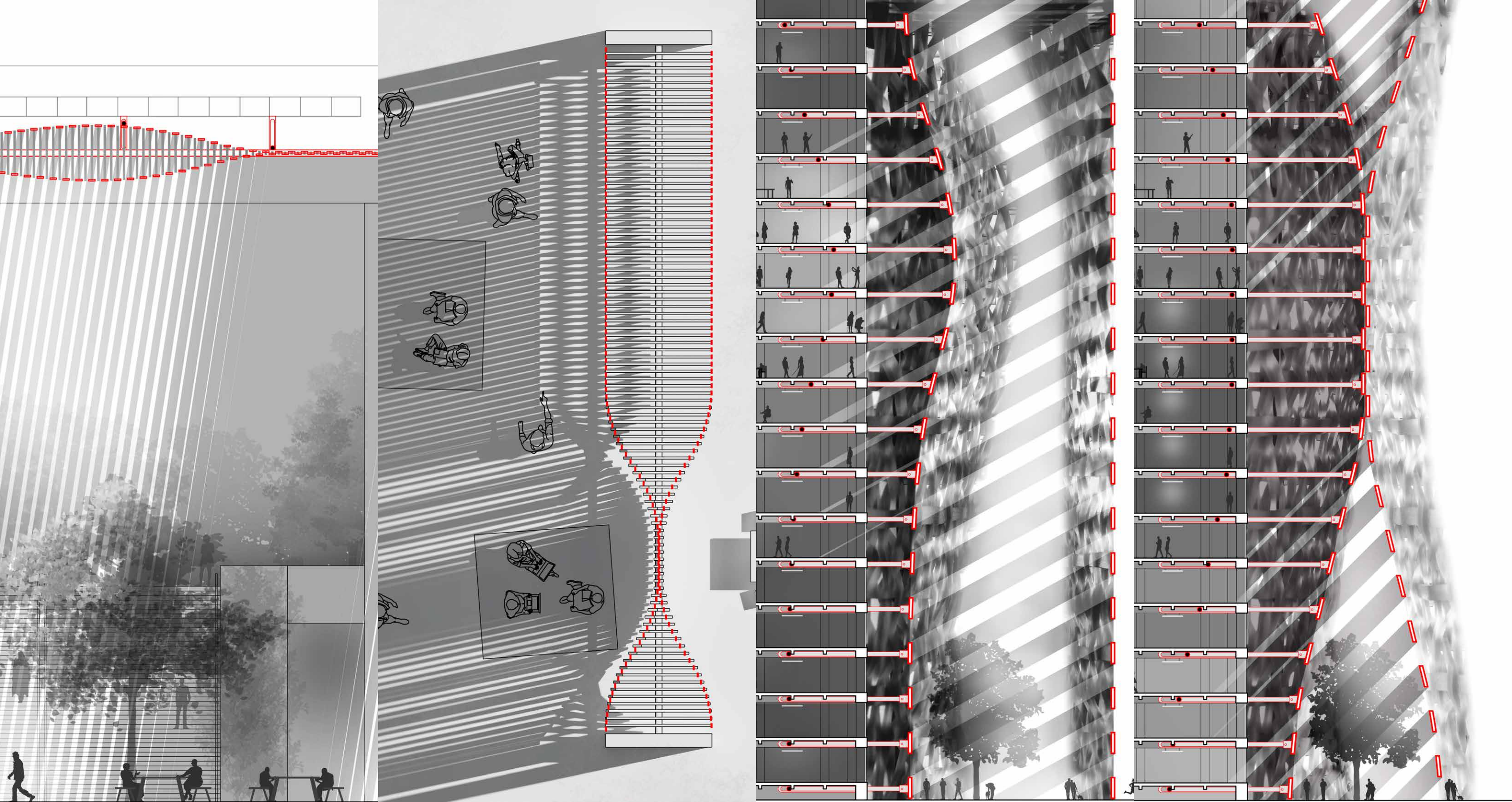




03.Transformable Sun Shade

Sun Shading Device with Plant-Based Mechanisms

HKU Year 3 Semester 1 Architecture Studio
Instructor: Miho Hirabayashi
09.2021 - 12.2021

Biomimetic Design
Design Performance Analysis

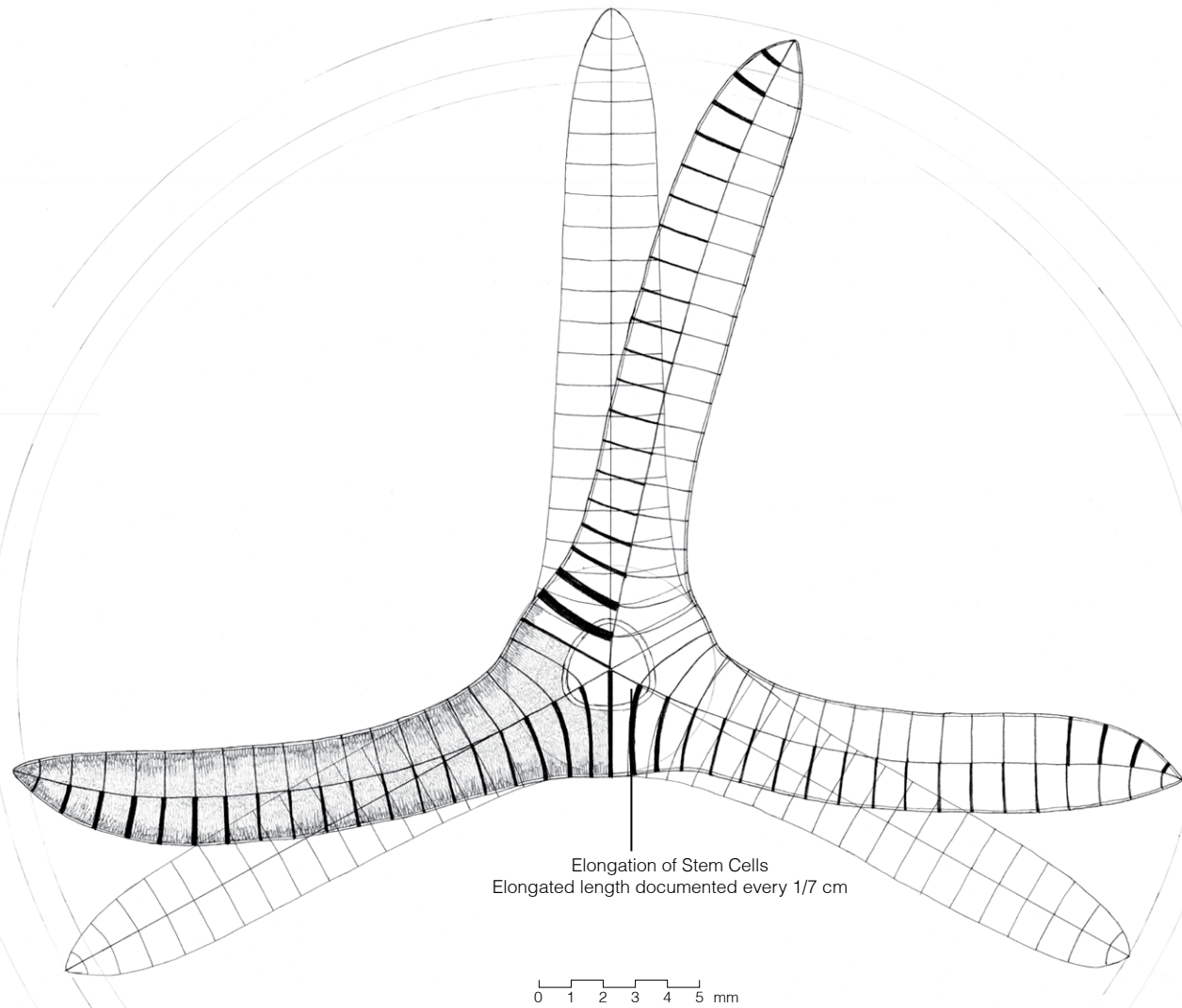
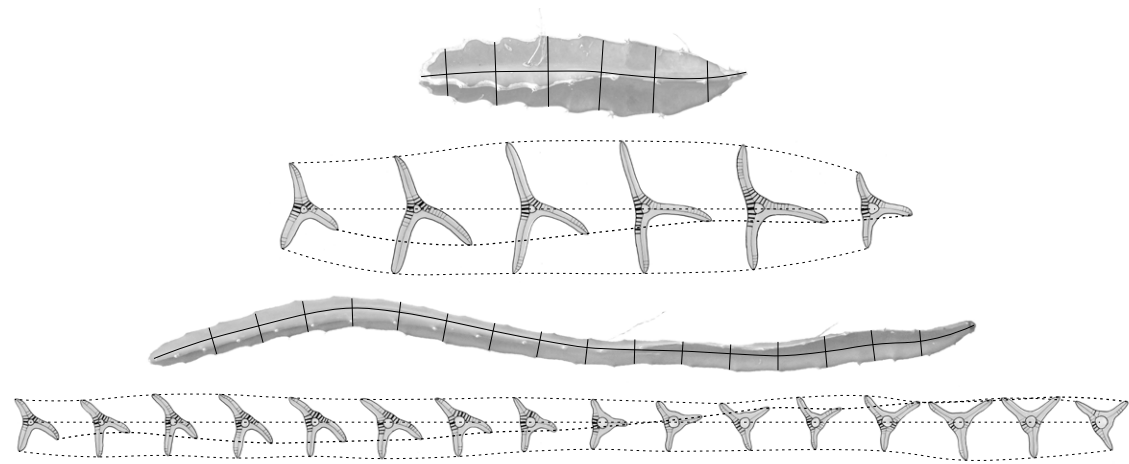
Tools:
Rhinoceros 3D
Vray

Unlike other architectural project, this studio is not initially constrained by program and site. Rather, the project departed from investigating the sun shading mechanism of the stem of dragon fruit. The idea is to learn from nature's mechanism, and to investigate how it can have architectural application.

In a dense tropical city like Hong Kong, both the climate and the density are the cause of increasing city temperature. Buildings are absorbing a strong intensity of solar radiation throughout the entire year. Some are absorbed into the building to heat up the indoor environment, causing more energy for mechanical ventilation, heating up the surrounding temperature even more. Reflected solar radiation is also a problem because the reflected solar radiation is

increasing the temperature of surrounding streets and buildings. The role of architectural shading devices in the dense city is becoming more critical to the thermal environment to the city.

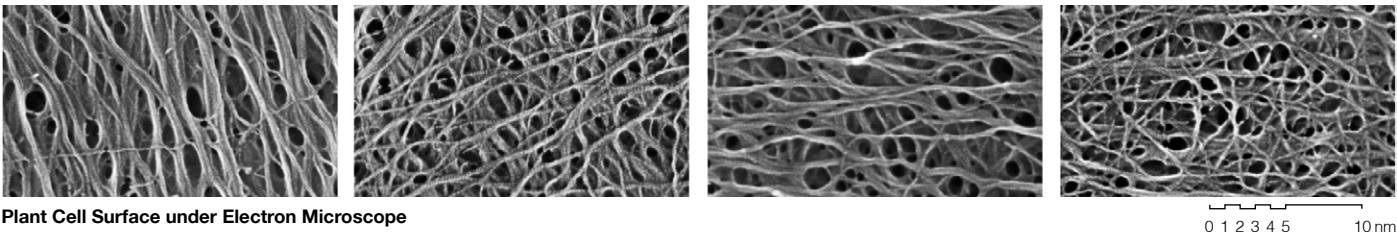
Buildings should be able to control unintended radiation in the city, and use them as a tool instead of an unsustainable factor to the people in the city. People under the shading devices will be able to observe the changing lighting quality very frequently. The constantly changing spaces can be easily visualized, enriching the environment of the city.



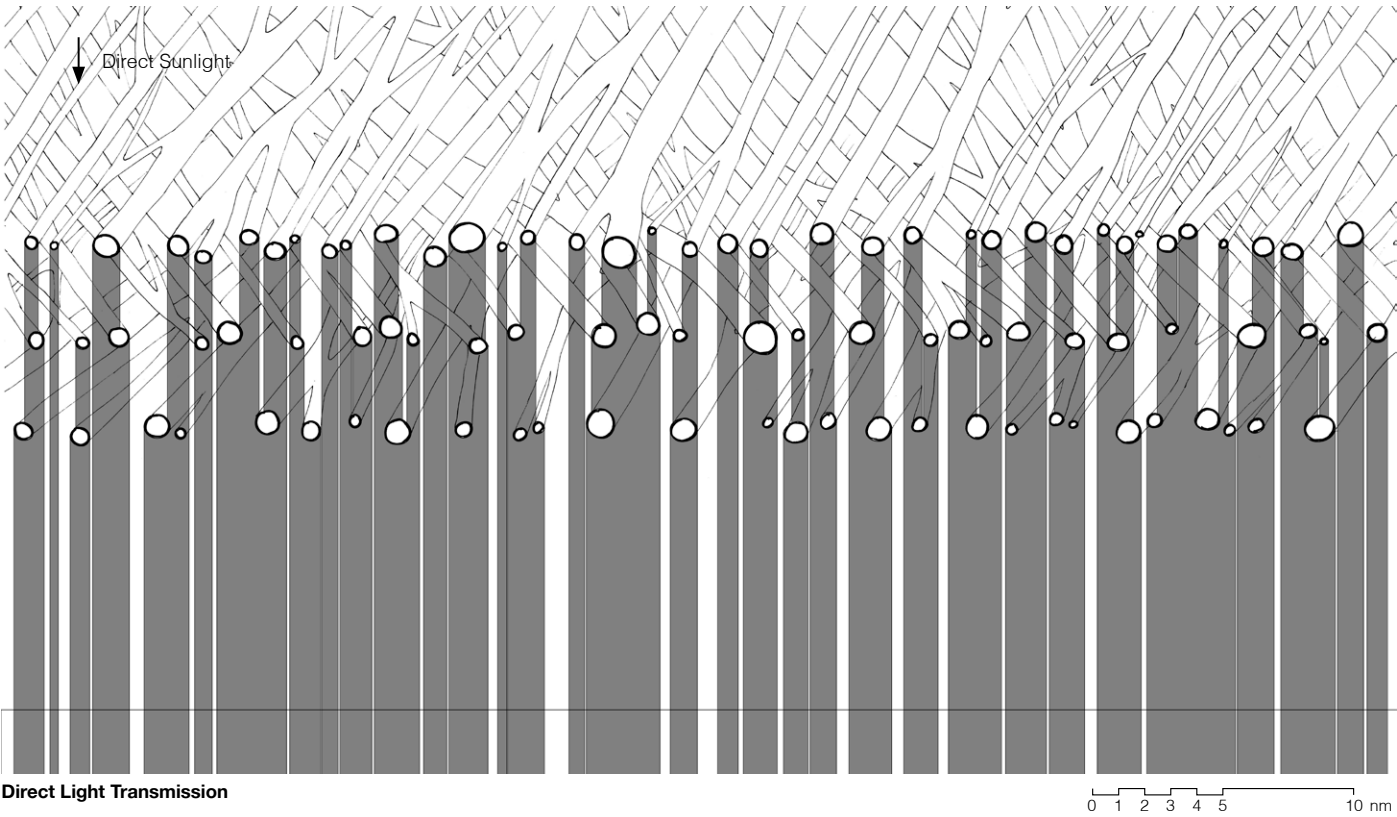
Dragon Fruit Stem Rotating Mechanism

Dragon fruit has unique mechanism to sunlight. Stem in the shaded area gets thinner and longer. It is because when dragon fruit is under the shade, more amount of Auxin is produces, the growth hormone responsible for elongating the plant cells. Longer stem means there is a higher chance of getting exposed to the sun. Therefore, more resource is distributed to the growth in terms of its length compared to the thickness.

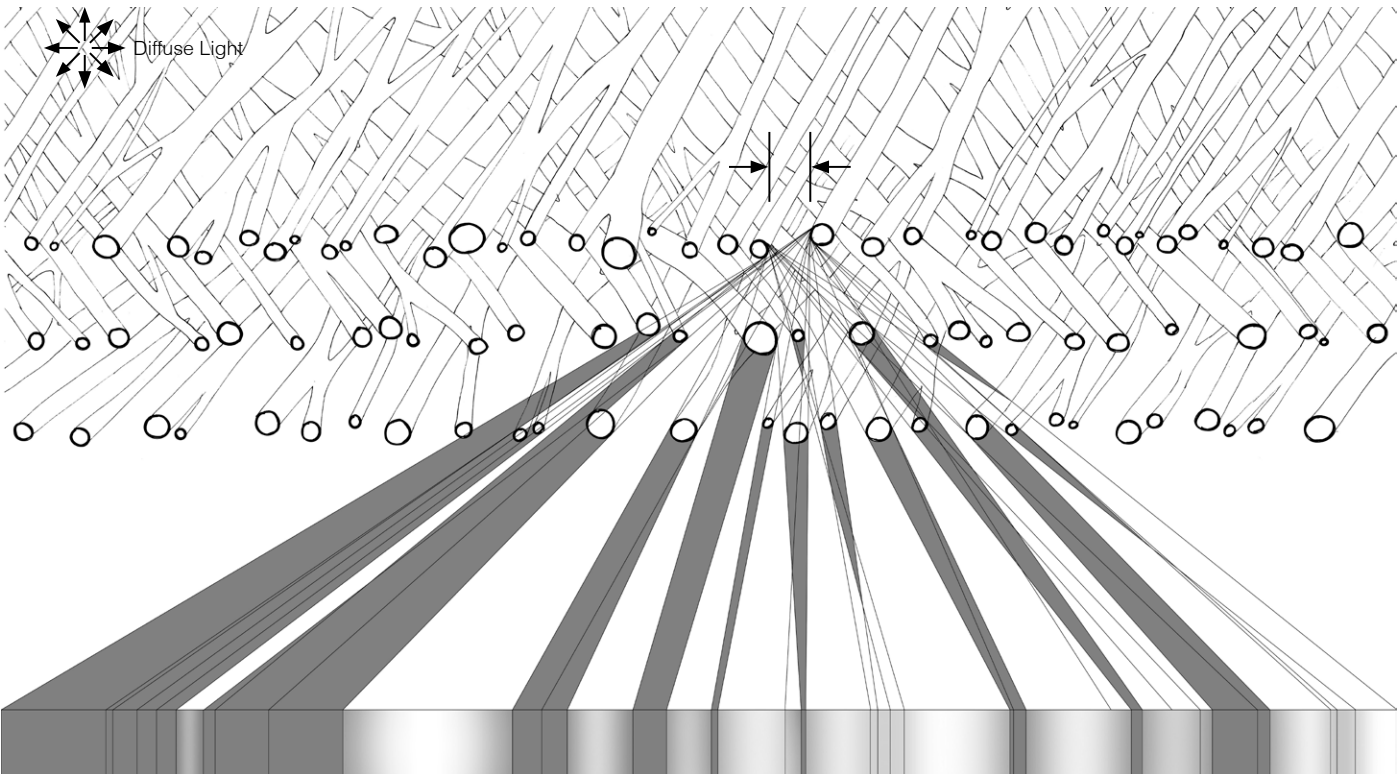
Same mechanism can be observed when looked closer into the cross section of the stem. The stem is composed of 3 wings extruding out from the vascular core. They will rotate toward the sunnier side of the stem when the shaded side is elongated to change the direction of the wings.



Plant Cell Surface under Electron Microscope

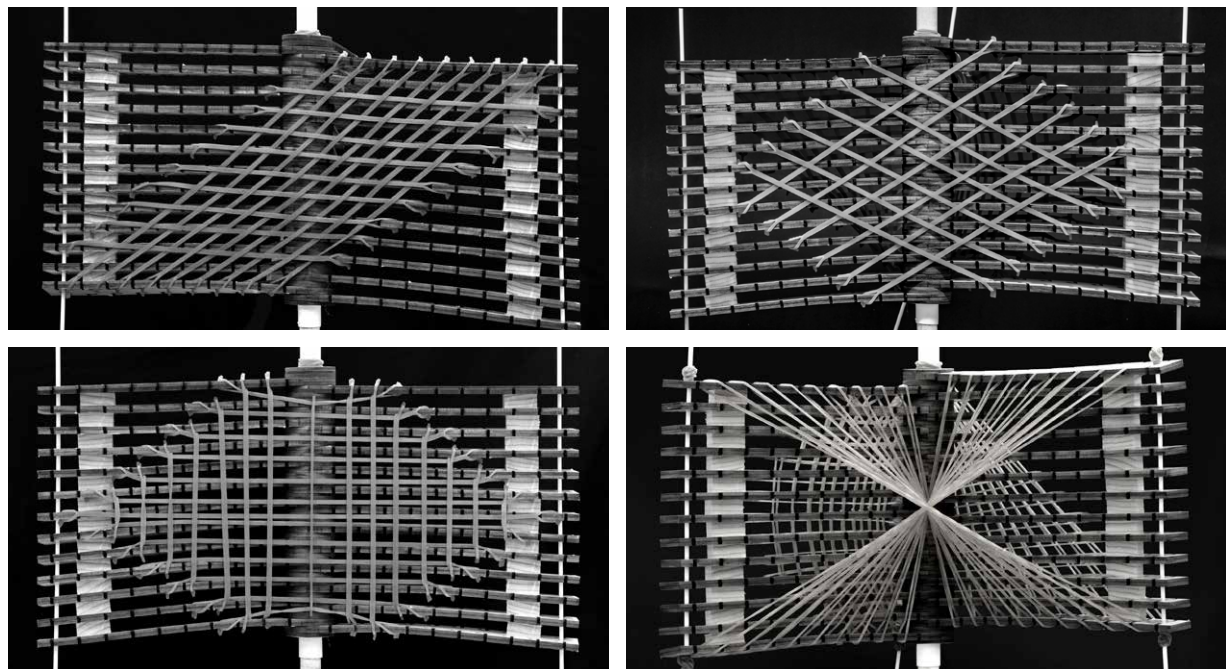
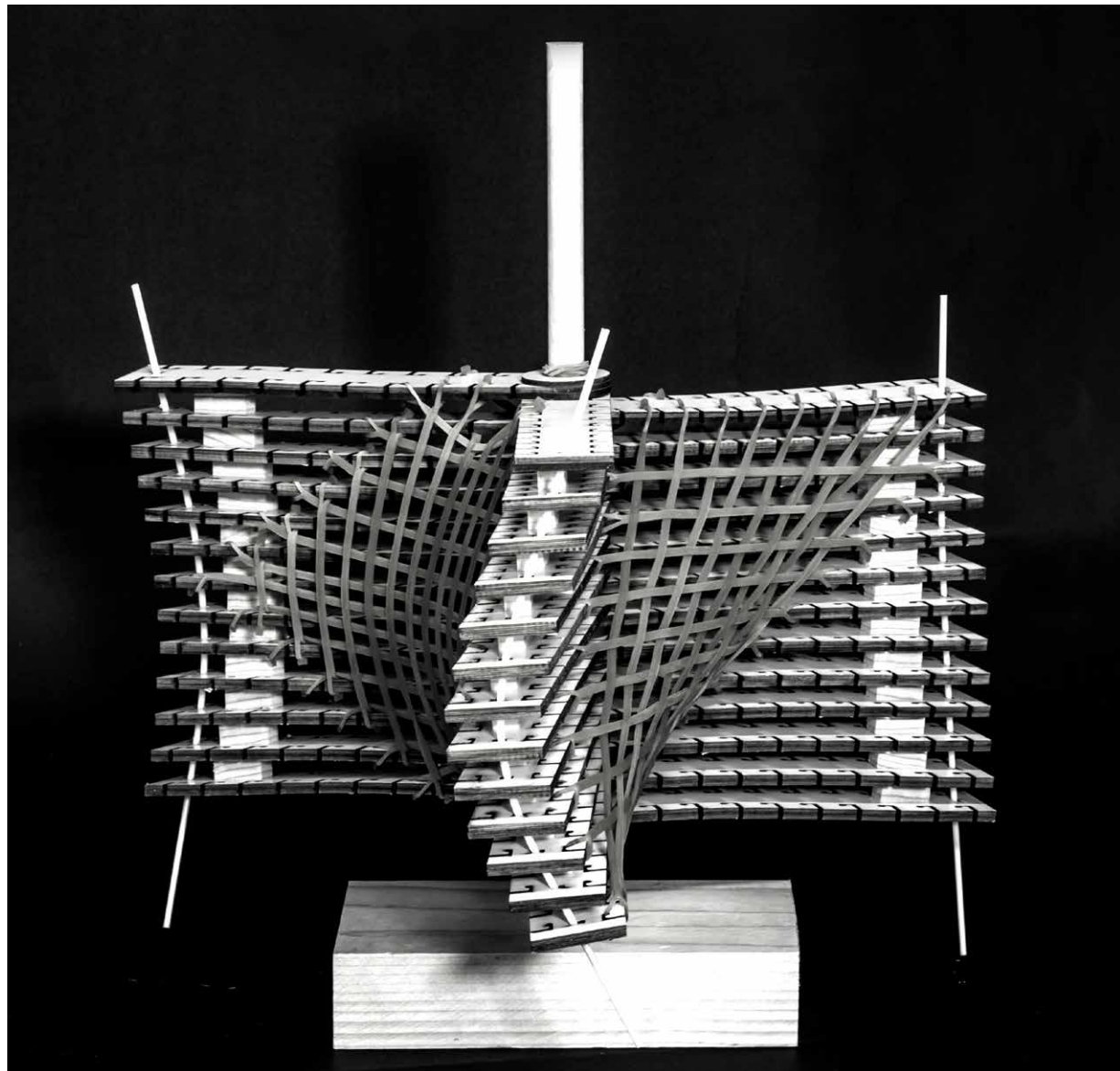


Direct Light Transmission



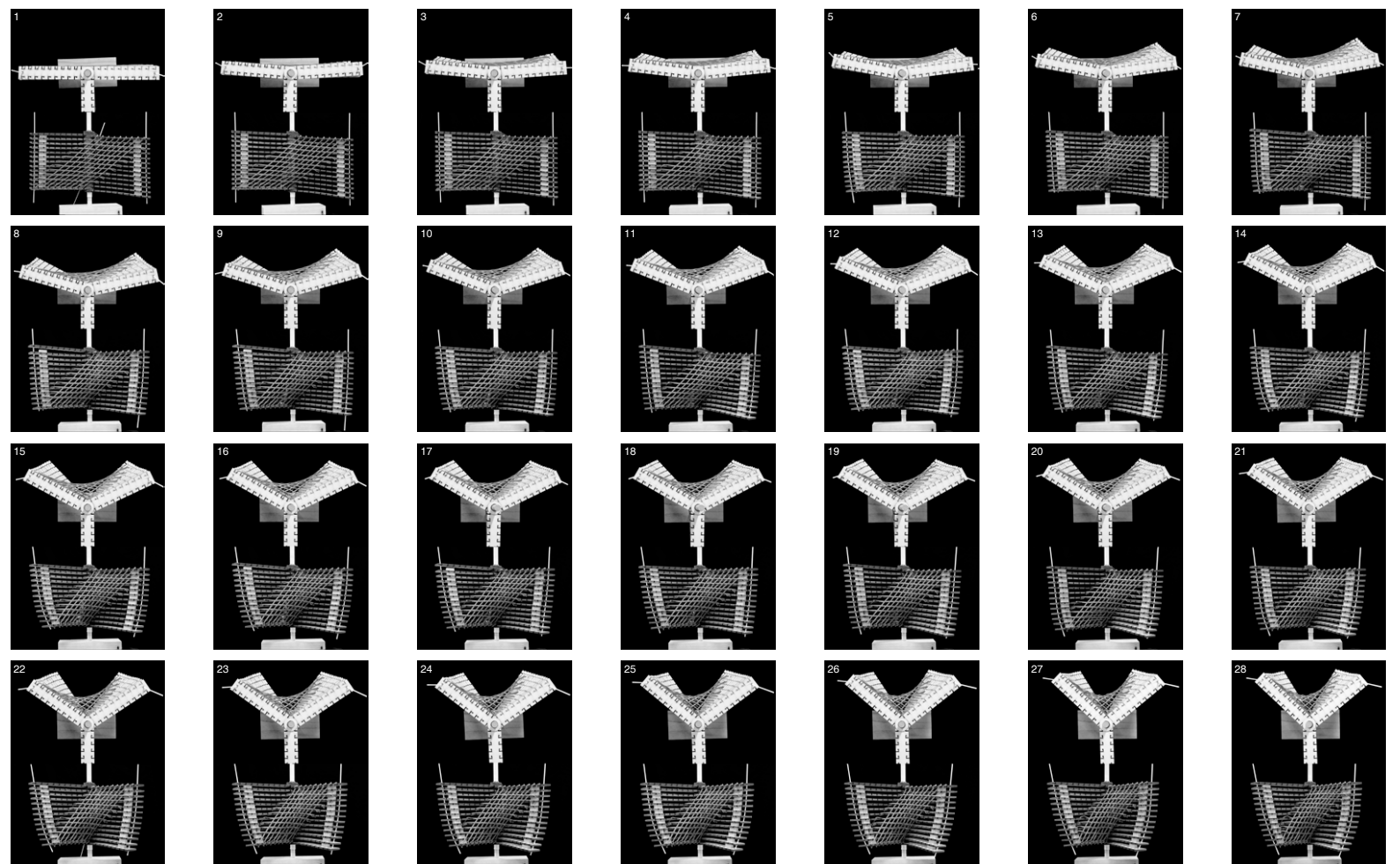
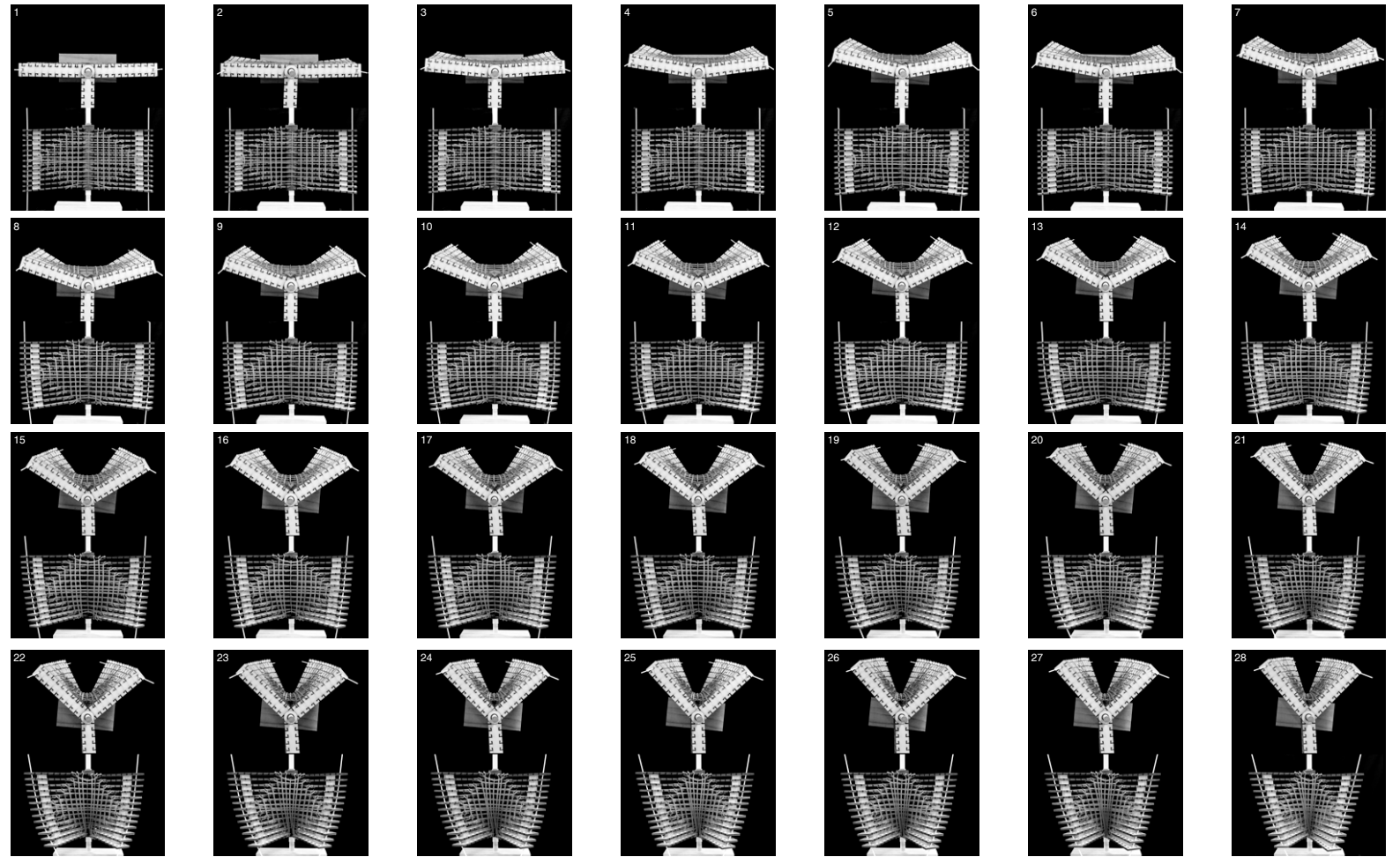
Diffuse Light Transmission through Selected Gap

The cellulose fiber at the surface of the cell acts as the sunlight filter. Light behaves differently when comparing direct light and diffuse light.



Surface Transform Test Model

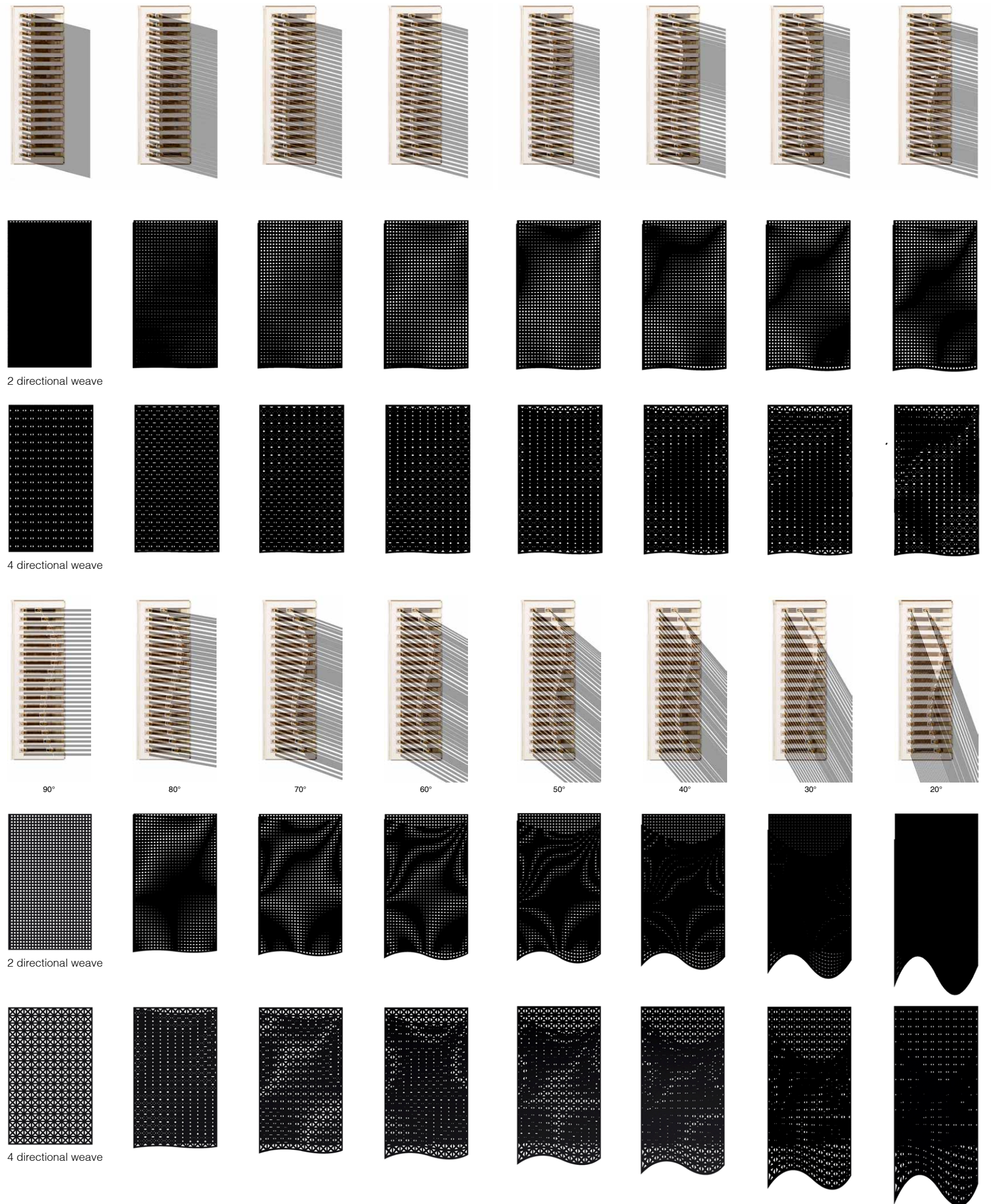
In the second testing model, the identical mechanism is shown in an enlarged model, the surface imitating the dragon fruit's skin is attached to the model. Multiple variation of surface is being tested. The surface tension and geometry changes as the model contracts and expands. The followings are the documentation of the changes for the 2 selected surface types.





Light Filtering Concept & Shadow Pattern Generation

The model is consisted of two layers of rubber band attached to the side of the rectangular frame. The surface is freely movable through individual movement of the knots. The distance between the two surfaces creates different porosity of the light in two different sides. 3 factors affect the behavior of light transmission: The distance between two surfaces, the angle of the incident light, and the geometry of the rubberband weave.

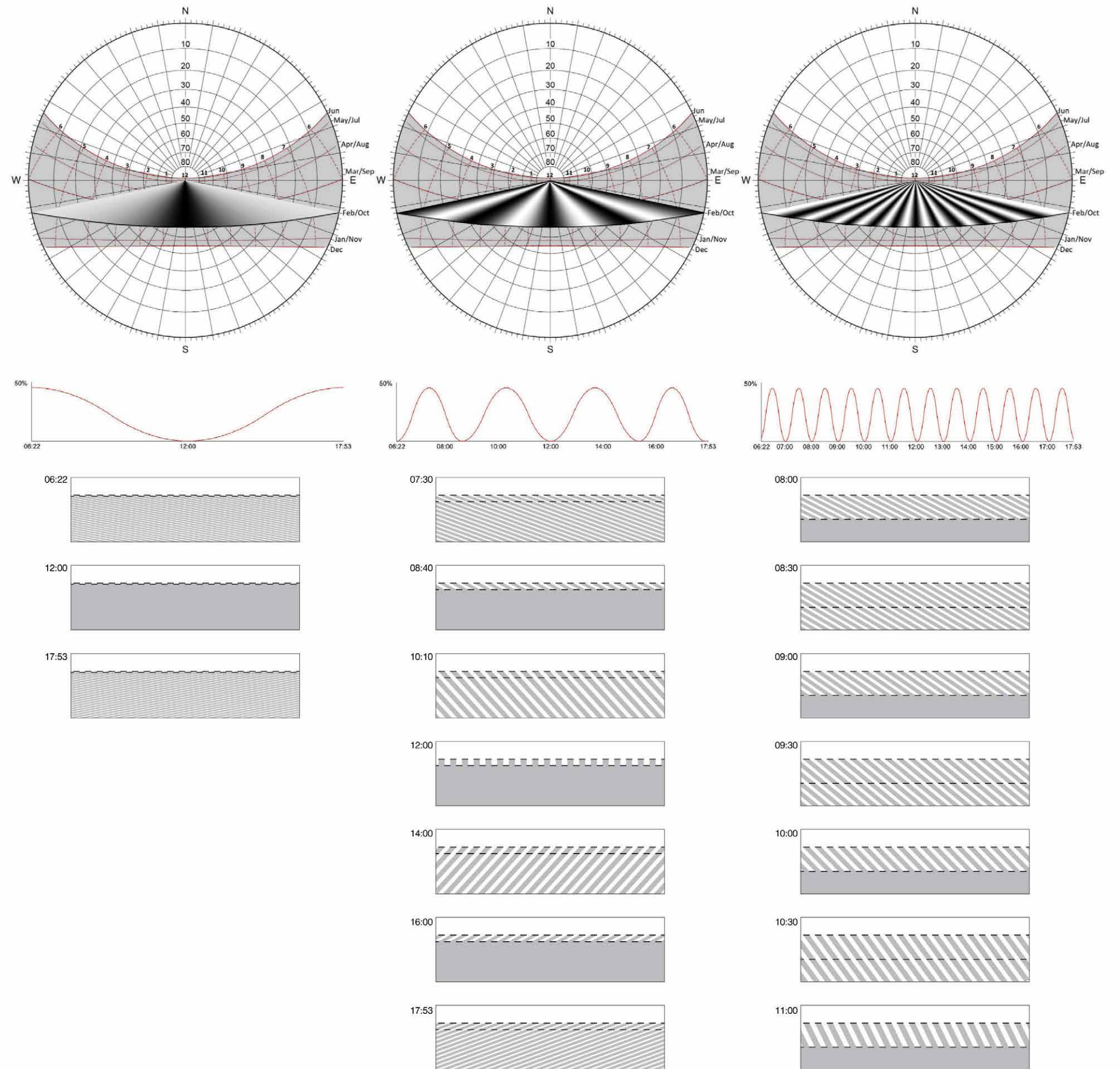


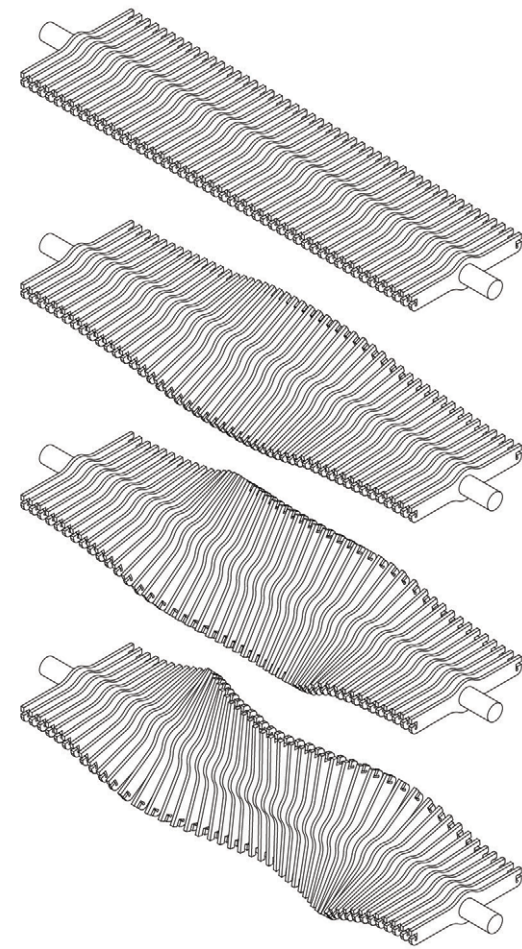
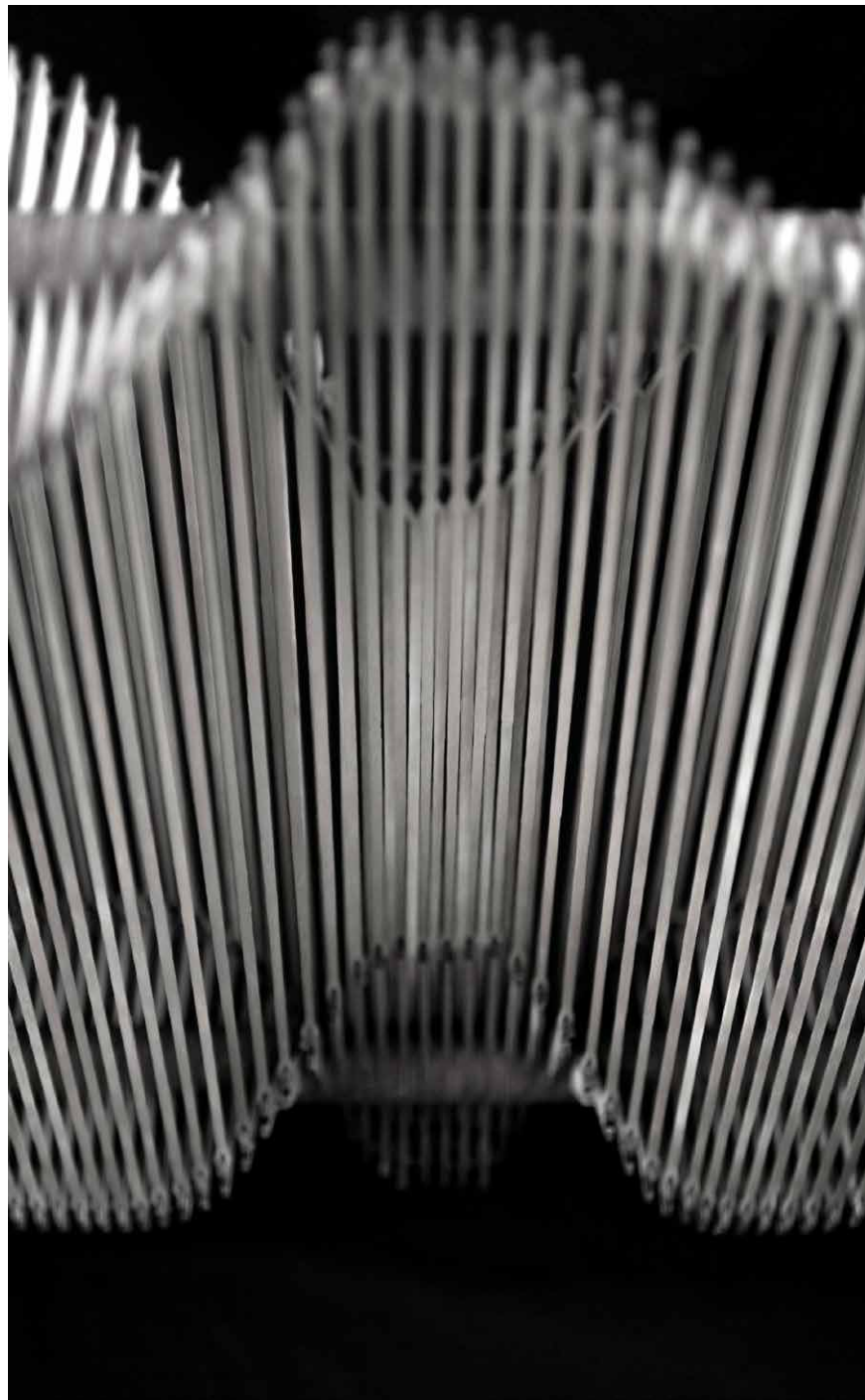


Testing model - Double Surface Transformation 2

The model attempts to overcome the limitations of the previous test model. The transformation mechanism is simplified where the distance between two surfaces is controlled by rotating the wings top and bottom.

When wing is rotated 90° to be completely vertical, the rubberband overlaps. Since the location of the rubberband completely compliments each other, the light is blocked completely.

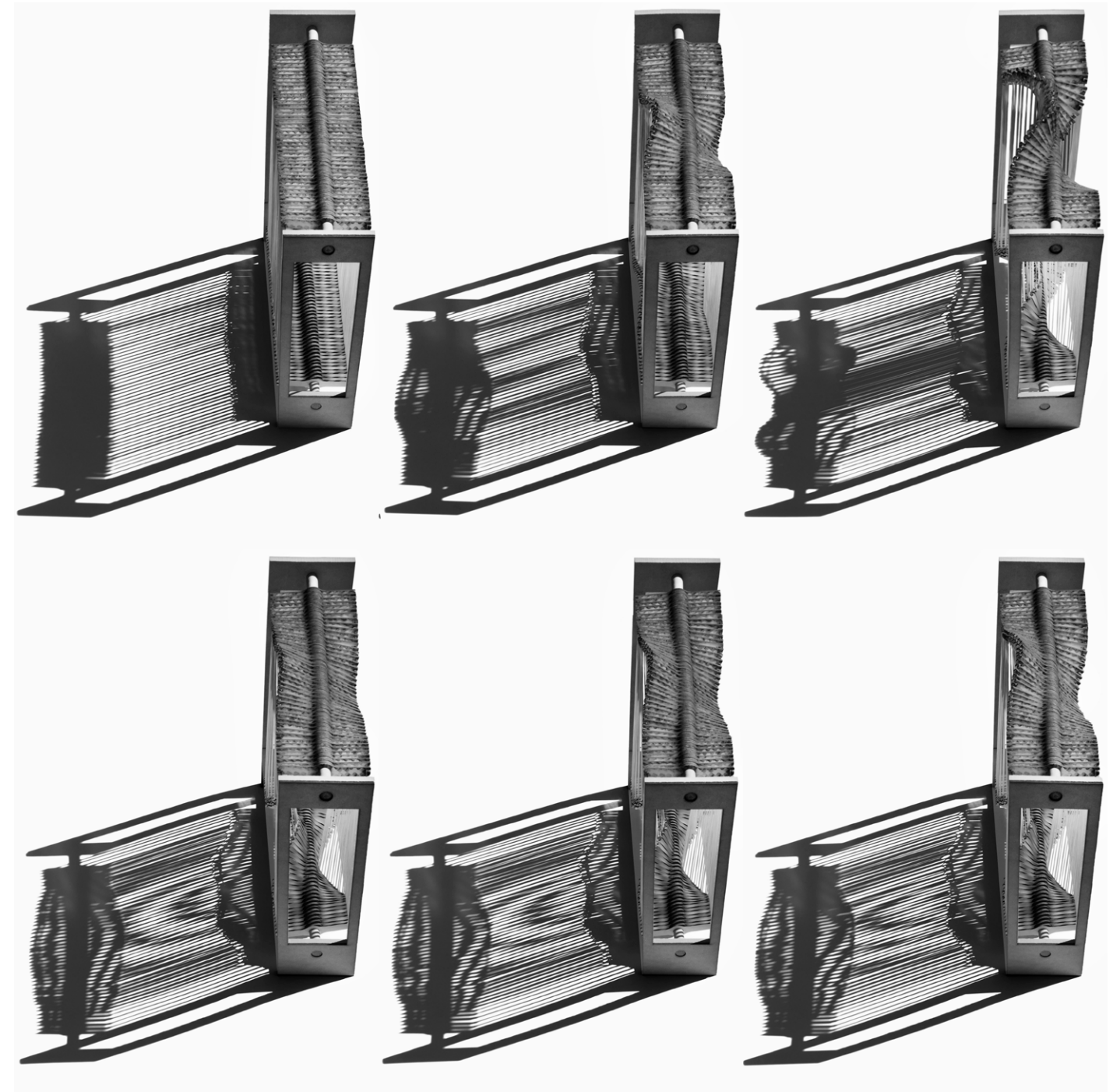




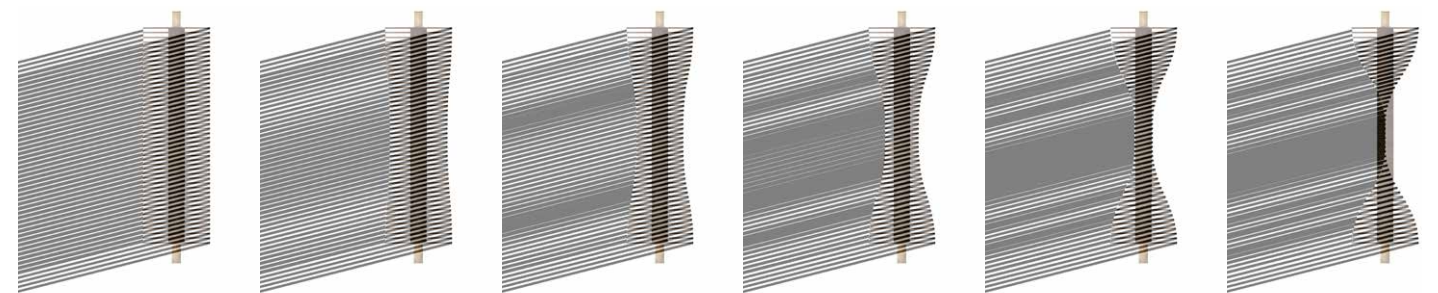
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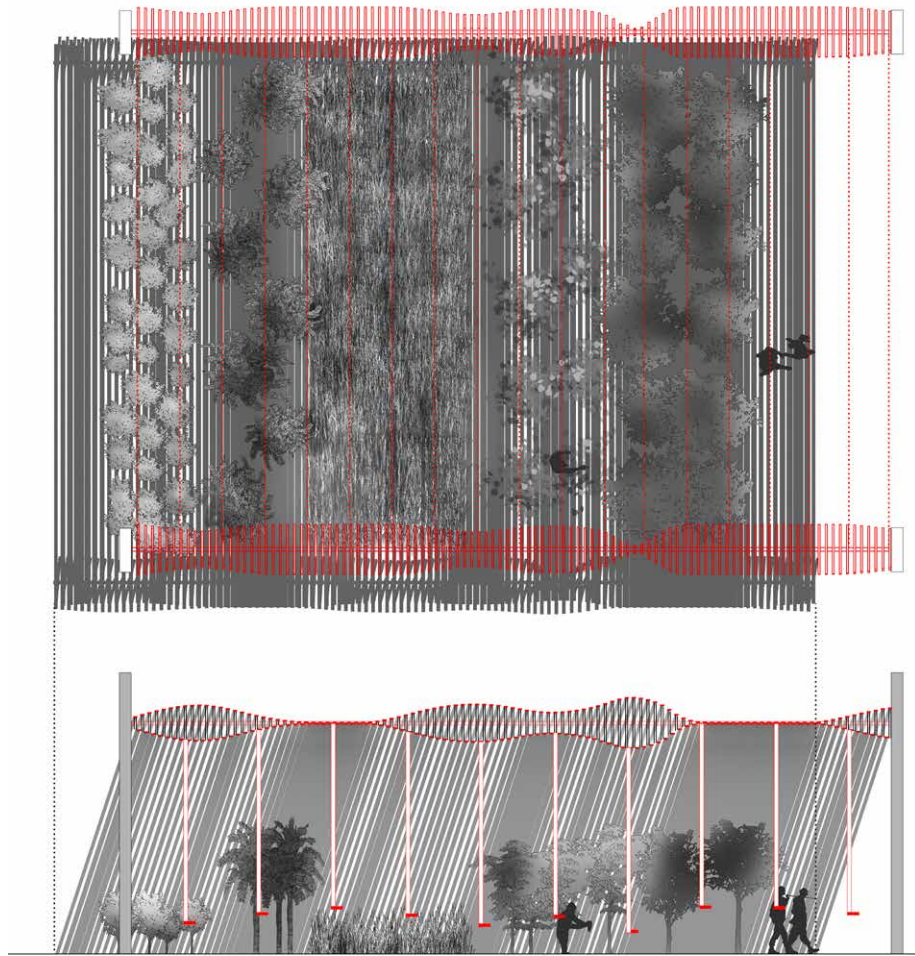
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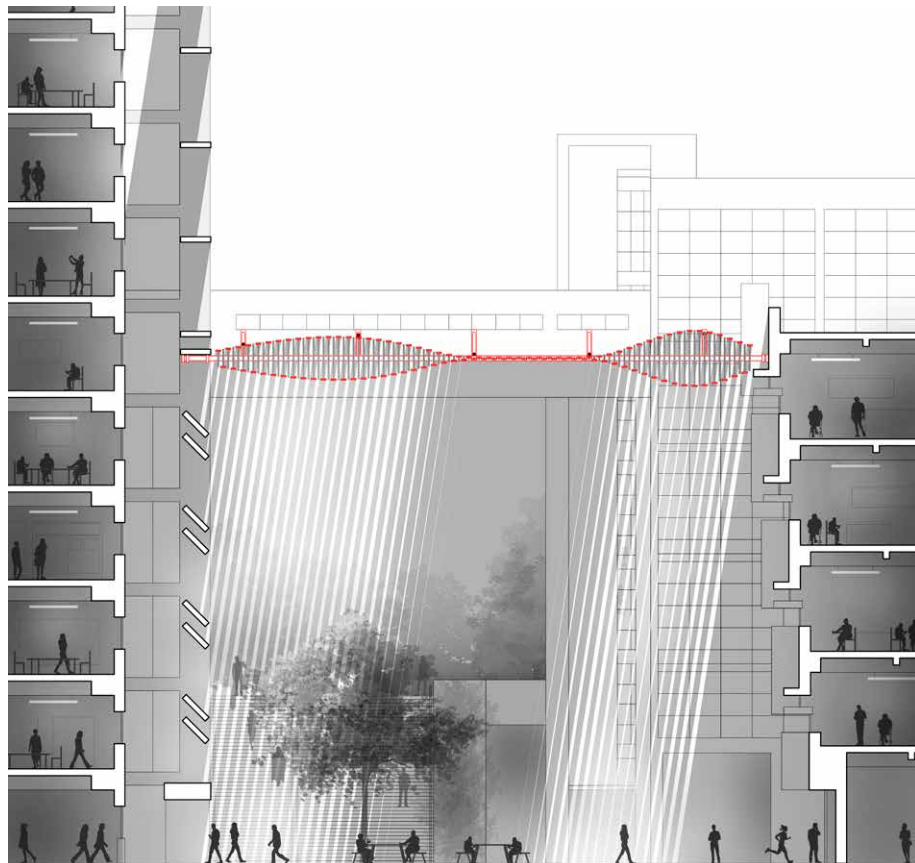
Shadow Documentation



Porosity Change Based on Wings' Rotation



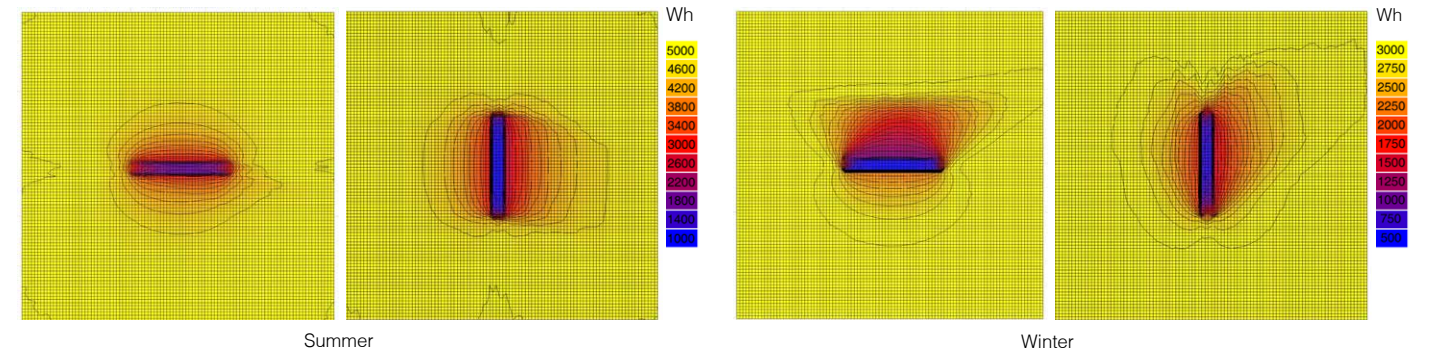
Context Type 1: Open Field
Arbitrary Farmland
Device Orientation: Horizontal



Context Type 2: Urban
Knowles Building, HKU, Hong Kong
Device Orientation: Horizontal

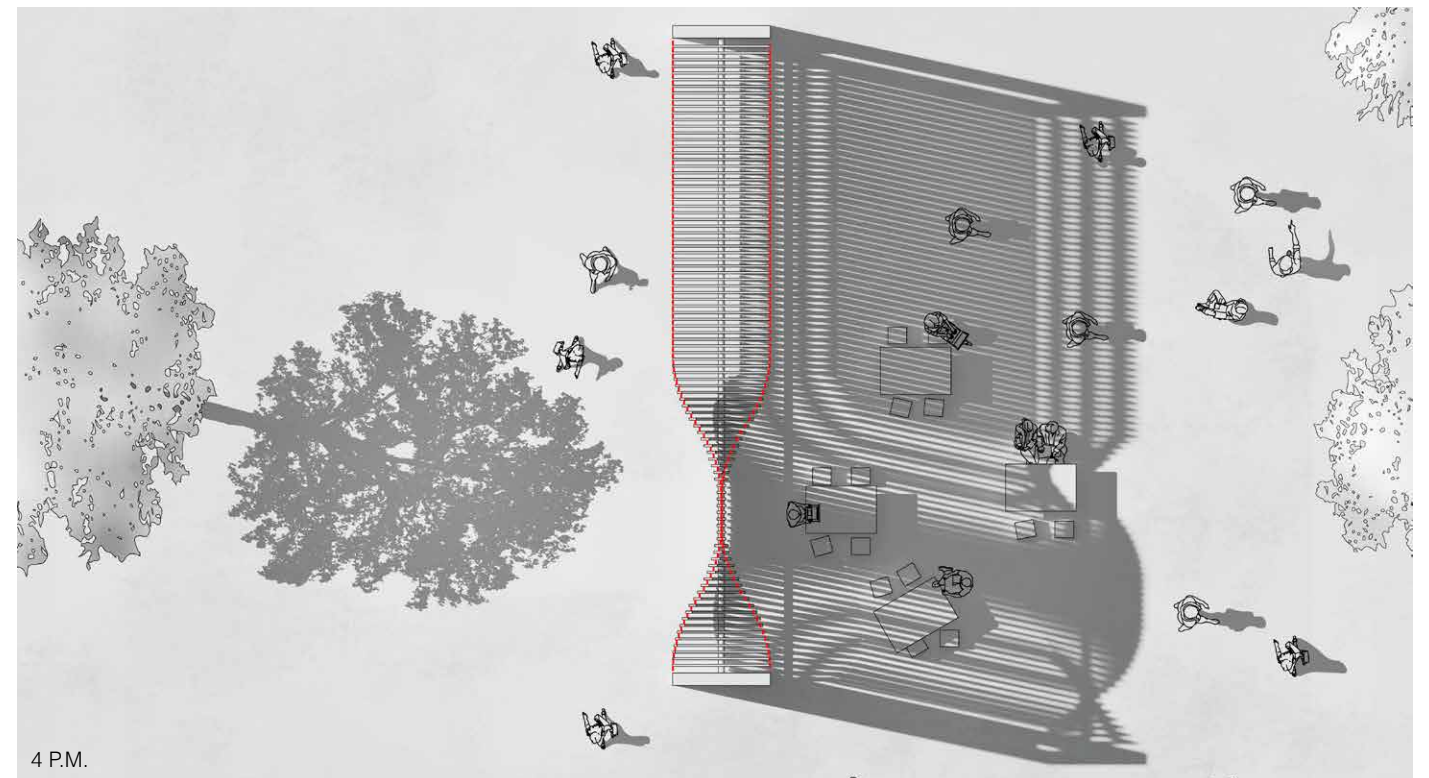
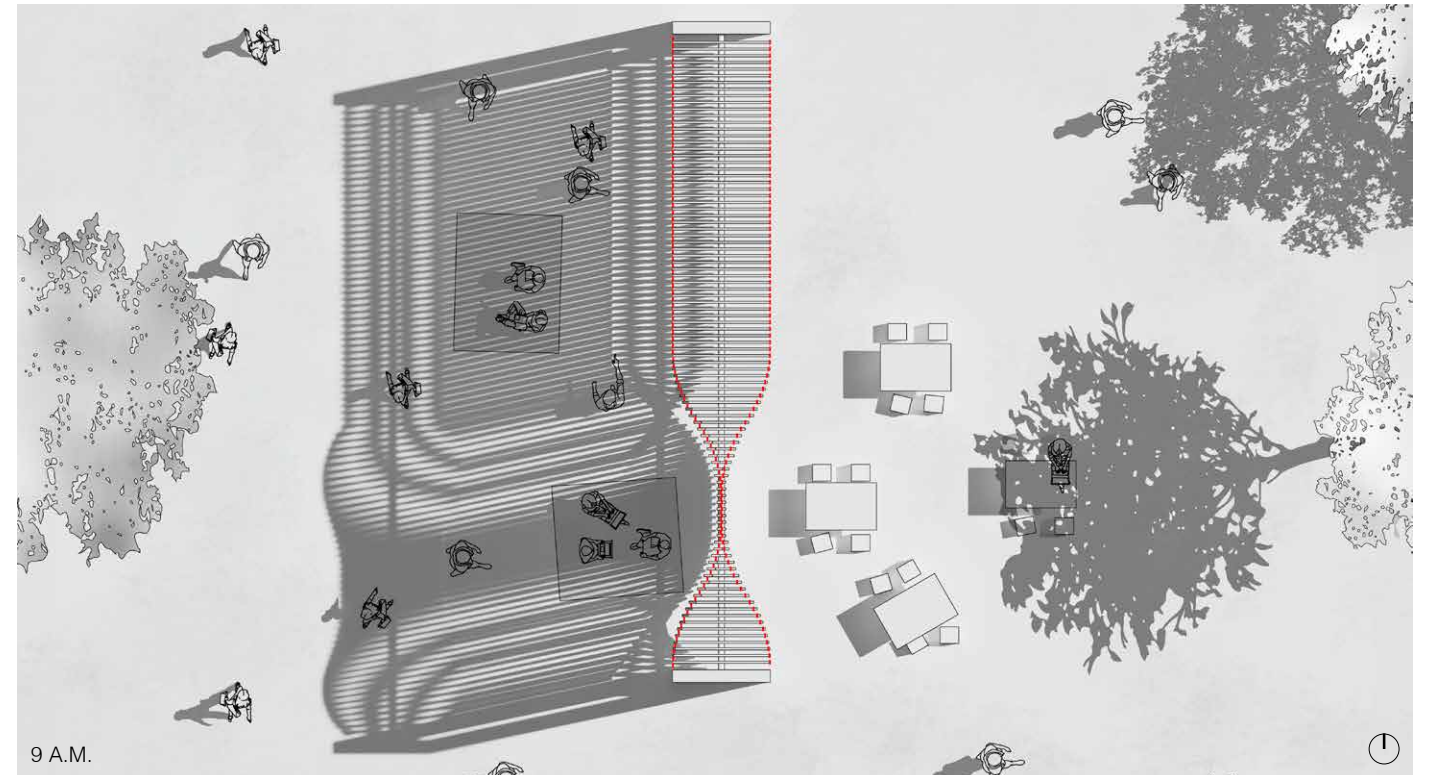
The testing models are placed in 2 different context: open field area & dense urban area. For each context, the device can be placed either vertically or horizontally. The context becomes more complex when the device placed vertically. Cardinal direction and the seasonal changes affect the effectiveness of the device. The effectiveness is more significant for winter seasons when the sun is low enough to show the have differences between cardinal direction

In an open field, a vertical shade will create shadows in all different orientations as the sun direction changes over the days. There will be almost no shadow at noon during the summer when the sun is almost completely 90 degree above in Hong Kong. The device has the potential of giving people the freedom of changing the intensity and location of the shadow and light

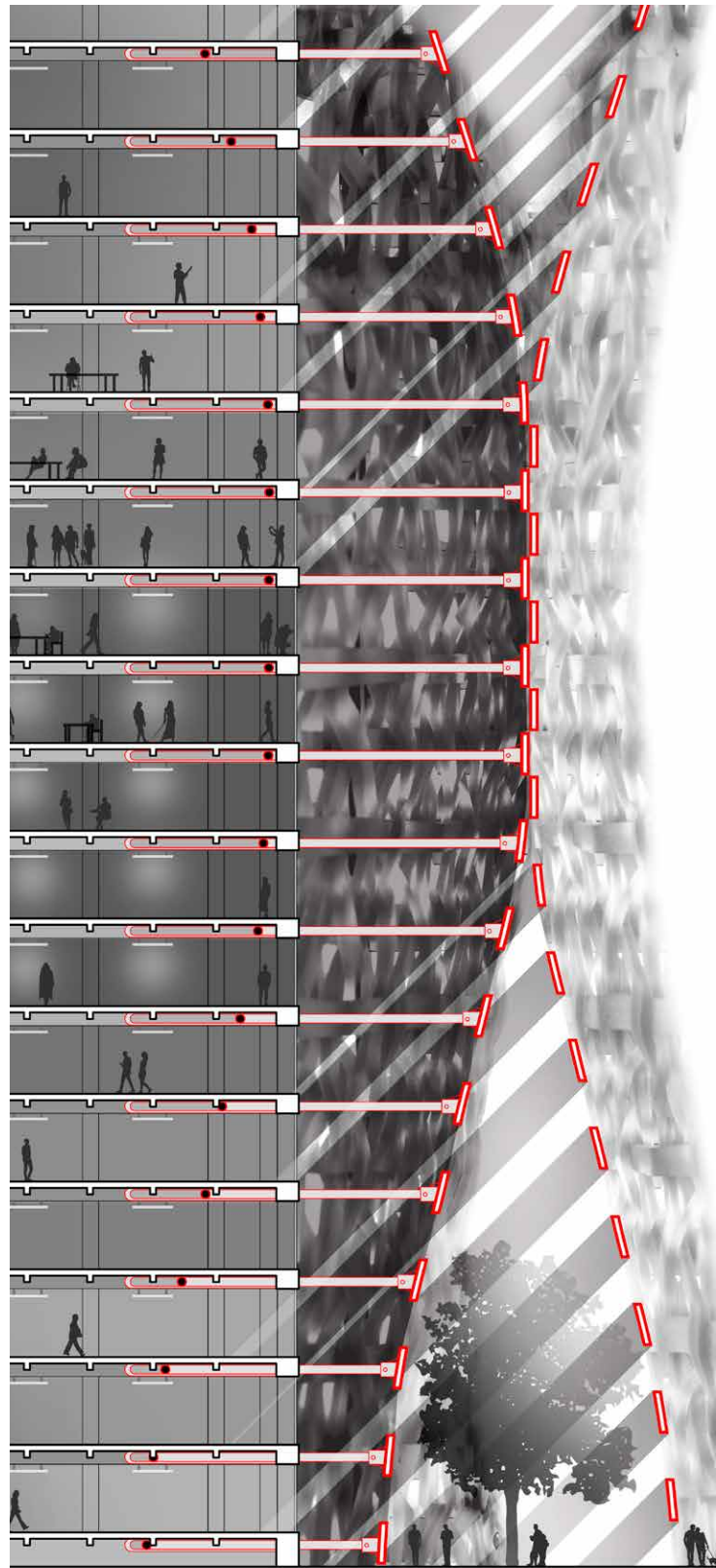


Summer

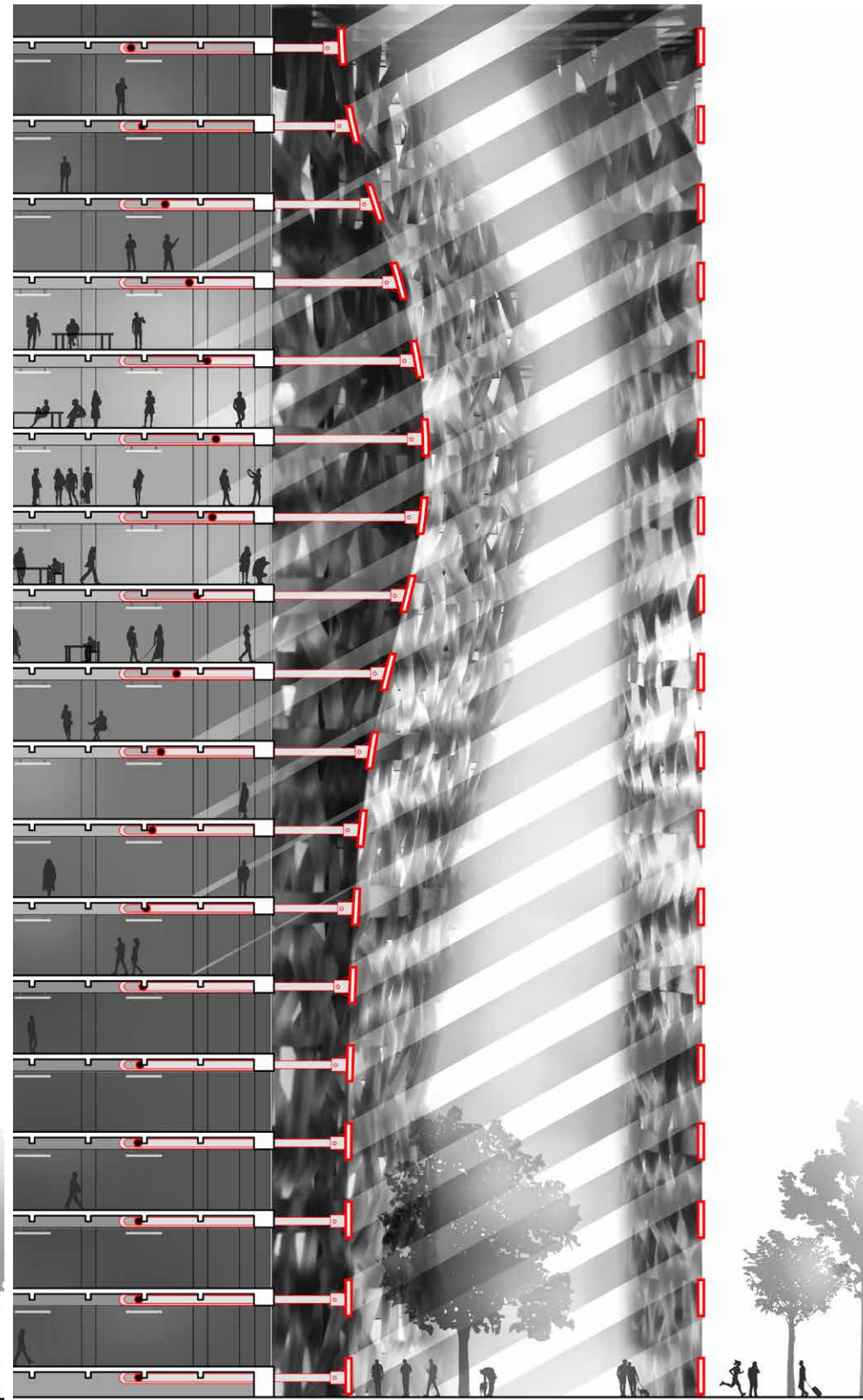
Winter



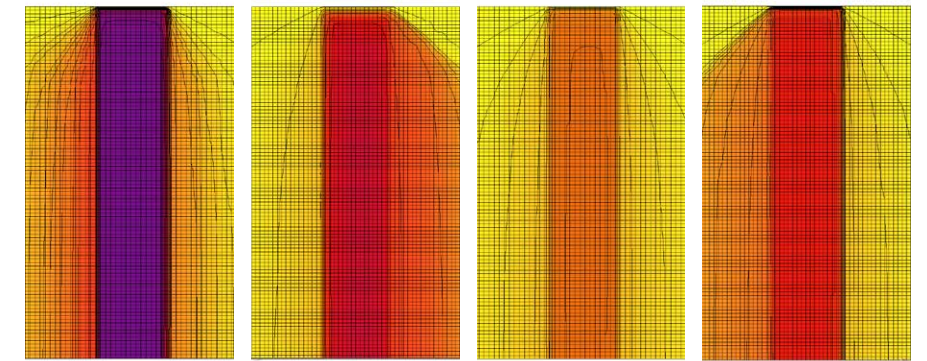
Context Type 3: Open Field
Arbitrary Park
Device Orientation: Vertical



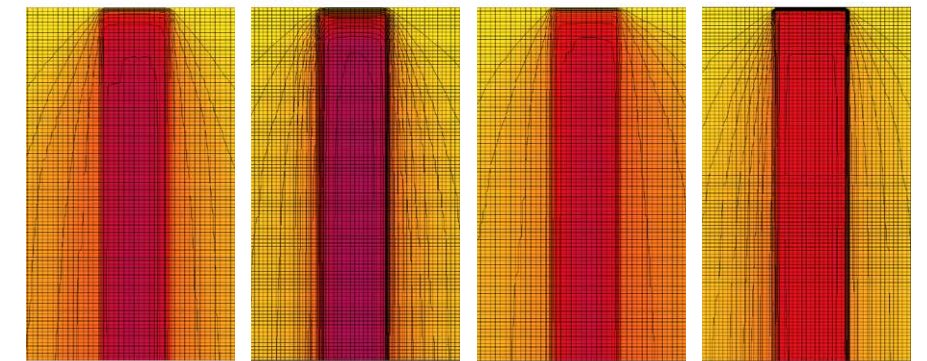
Context Type 4-2: Urban
Arbitrary Highrise Building
Device Orientation: Vertical



Context Type 4-1: Urban
Arbitrary Highrise Building
Device Orientation: Vertical



Winter



Summer

Daily Average Solar Radiation Analysis on Highrise Building Facade in Hong Kong

When the Device is attached to the highrise building vertically, three layers of space (outside, in between, inside) are created. The in between space is habitable as a visually dynamic space for various activities, while the surfacess can allow different amount of light to enter into different floors. The surface can be changed mechanically.